

star map generator notes, mostly for nothing- i just wanna type

1 theory

ok so the whole project is basically one question dressed up as a python script: given a star's fixed spot in the sky, and where i am + what time it is, where is that star right now.

so there's two coordinate systems having a little disagreement here.

equatorial coordinates (RA/dec) — this is how the catalog stores stars, basically lat/long but for the sky.

RA is like longitude except measured in hours instead of degrees because apparently old astronomers looked at 360° , a perfectly normal number and said non merci, we're doing clocks instead.

declination is like latitude, -90° to $+90^\circ$, much more reasonable behaviour. since stars are stupidly far away, these barely budge over a human lifetime — so one RA/dec pair per star, forever et catalog done.

since $24\text{h} = 360^\circ$,

$$1^{\text{h}} = 15^\circ \tag{1}$$

so converting RA hours $\rightarrow$ degrees is just times 15. i will forget this by tuesday, hence it is now written down forever.

horizontal coordinates (alt/az) — this one i actually care about because it gets plotted. altitude = how high up (0° = horizon, 90° = directly overhead, negative = below the horizon = don't bother looking, it's not there). azimuth = compass direction, 0° north, 90° east blah blah.

annoying part: this depends on where you're standing AND what time it is, because earth is rude and keeps spinning without asking anyone.